Lengths and ratios

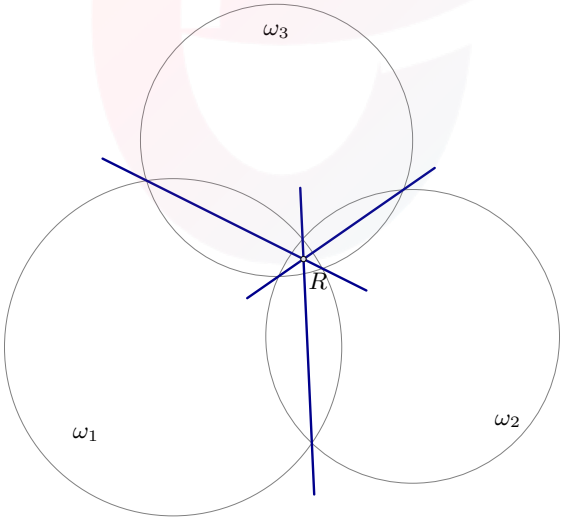
L4-GE2 Level 4 Olympiad · Geometry

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Power of a point, radical axes, the ratio lemma with Ceva and Menelaus, and mass points, for AIME computation and BMO proofs.
By the end you can use signed power of a point, radical axes and radical centres in proofs and computations; turn angle data into side ratios with the ratio lemma and trig Ceva; read off ratios and areas in cevian configurations with mass points.
Prerequisites: Power of a point and cyclic quadrilaterals (L3-GE1), Trigonometry and coordinates (L3-GE2), Olympiad angle chasing (L4-GE1). **Contests:** BMO1, MOG, AIME, USAJMO. **Time:** about 180 minutes.

1 Lesson

If three circles meet in pairs, the three lines containing their common chords pass through one point. There is no useful angle to chase in this figure, and the proof below uses the *power of a point* instead. This unit covers the methods based on lengths and ratios: power of a point, radical axes, the ratio lemma with Ceva and Menelaus, and mass points. They are the standard methods for AIME geometry and for BMO1 problems about lengths.



1.1 Power of a point

Theorem.
Let ω have centre O and radius r, and let P be any point. If a line through P meets ω at A and B, then

$$\overline{PA} \cdot \overline{PB} = OP^2 - r^2,$$

where bars denote signed lengths along the line, so the product is negative exactly when P

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is inside the circle. The number $\text{pow}(P) = OP^2 - r^2$ is the *power* of P. If PT is a tangent, $PT^2 = \text{pow}(P)$.

Proof. let M be the foot of the perpendicular from O to the line, which is the midpoint of AB. With signed lengths, $\overline{PA} \cdot \overline{PB} = (\overline{PM} + \overline{MA})(\overline{PM} - \overline{MA}) = PM^2 - MA^2 = (OP^2 - OM^2) - (OA^2 - OM^2) = OP^2 - r^2$. The tangent is the case $A = B = T$.

The converse is used just as often. If lines AB and CD meet at P and $\overline{PA} \cdot \overline{PB} = \overline{PC} \cdot \overline{PD}$, then A, B, C, D are concyclic: line PC meets circle ABC again at a point D' with $\overline{PC} \cdot \overline{PD'} = \overline{PA} \cdot \overline{PB}$, so $D' = D$. Likewise, $PT^2 = \overline{PA} \cdot \overline{PB}$ makes PT tangent to circle ABT.

1.2 Radical axis and radical centre

Radical axis.
For two circles with different centres, the points with equal power with respect to both form a line perpendicular to the line of centres. If the circles meet at X and Y, it is line XY; if they touch, it is the common tangent at the point of contact.
Radical centre. For three circles with non-collinear centres, the three radical axes meet at one point.

Put the centres at $(0,0)$ and $(d,0)$. The condition $x^2 + y^2 - r_1^2 = (x-d)^2 + y^2 - r_2^2$ simplifies to $x = \dfrac{d^2 + r_1^2 - r_2^2}{2d}$, a line perpendicular to the line of centres, and common points of the circles (power 0 for both) lie on it. For three circles, two of the axes are perpendicular to non-parallel lines, so they meet at a point R; then R has the same power for all three circles, so it lies on the third axis. This proves the statement at the start of the unit.

In practice, a point on the common chord line has equal tangents to both circles; four points are concyclic when two lines through them meet at a point with equal products; three lines are concurrent when they are radical axes of three circles.

1.3 The ratio lemma, Ceva and Menelaus

Ratio lemma.
If D lies on side BC of triangle ABC, then

$$\frac{BD}{DC} = \frac{AB \sin \angle BAD}{AC \sin \angle DAC}.$$

Ceva. For D, E, F on sides BC, CA, AB, the lines AD, BE, CF are concurrent if and only if $\dfrac{BD}{DC} \cdot \dfrac{CE}{EA} \cdot \dfrac{AF}{FB} = 1$. In trigonometric form, if and only if $\dfrac{\sin \angle BAD}{\sin \angle DAC} \cdot \dfrac{\sin \angle CBE}{\sin \angle EBA} \cdot \dfrac{\sin \angle ACF}{\sin \angle FCB} = 1$.
Menelaus. If a line meets lines BC, CA, AB at D, E, F, the same product of unsigned ratios equals 1; conversely, if the product is 1 and one or three of the points lie outside the sides, the points are collinear.

The ratio lemma is an area comparison: triangles ABD and ADC share the height from A, so

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$\frac{BD}{DC} = \frac{[ABD]}{[ADC]} = \frac{\frac{1}{2}AB \cdot AD \sin\angle BAD}{\frac{1}{2}AC \cdot AD \sin\angle DAC}$. The angle bisector theorem is the special case of equal angles. For Ceva, if the cevians meet at P then $BD/DC = [ABP]/[ACP]$ (subtract the two pairs of triangles with bases BD, DC), and the three such ratios multiply to 1; the converse follows because the ratio determines the point. Trig Ceva follows from Ceva by applying the ratio lemma three times, since the side lengths cancel. Use the ratio lemma whenever you know angles at a vertex and want a ratio on the opposite side, or the reverse.

1.4 Mass points

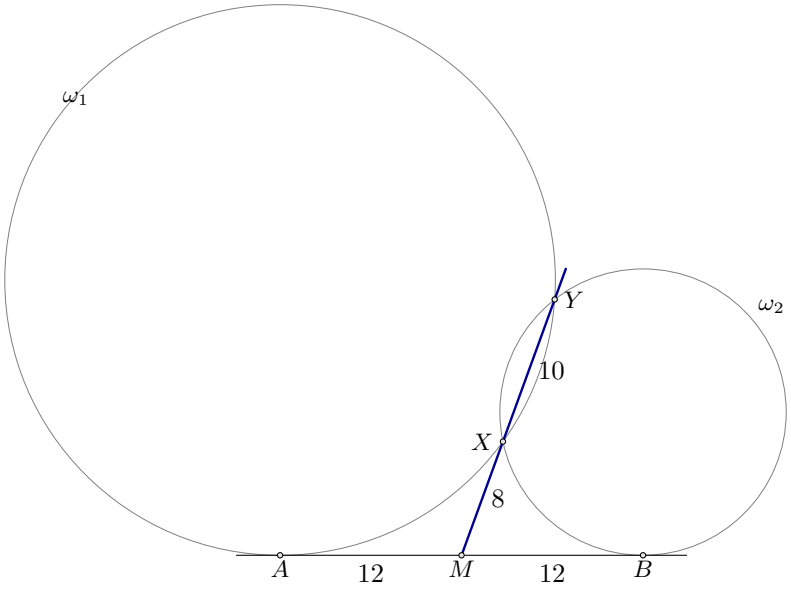
Mass points record the ratios in a Ceva configuration. Put masses $m_A, m_B, m_C > 0$ at the vertices. The centre of mass of B and C is the point D of BC with $BD : DC = m_C : m_B$ (the lever law: the heavier mass is nearer), and it carries mass $m_B + m_C$. The centre of mass $P = \dfrac{m_A A + m_B B + m_C C}{m_A + m_B + m_C}$ of all three is also the centre of mass of A and D, so P lies on AD with $AP : PD = (m_B + m_C) : m_A$. Similarly, grouping the masses differently shows P is on the other two cevians as well. Moreover $[PBC] : [PCA] : [PAB] = m_A : m_B : m_C$. So once the masses match two given ratios, every other ratio and area in the figure can be read off.

1.5 Worked examples

Example 1.
Circles ω_1 and ω_2 meet at X and Y. A common tangent, nearer to X than to Y, touches ω_1 at A and ω_2 at B. Given $AB = 24$ and $XY = 10$, find the distance from X to the midpoint of AB.

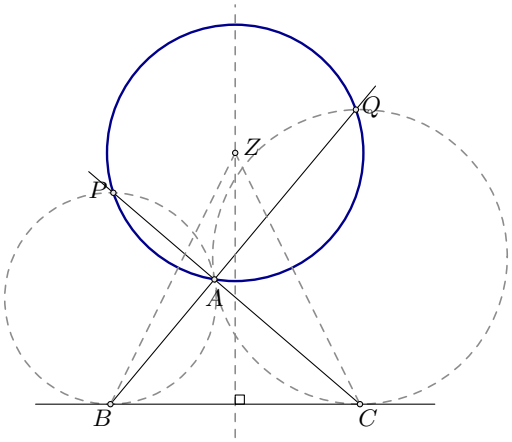
Solution.
Neither radius is known, so coordinates are awkward. The circles are related through their common chord, which lies on their radical axis, so we use powers. Let line XY meet AB at M. Since M is on the radical axis, $MA^2 = MX \cdot MY = MB^2$, so M is the midpoint of AB and $MA = 12$.
The tangent line lies outside both circles, so M has positive power and X, Y are on the same side of M, with X nearer. Hence, writing $MX = x$, we get $x(x + 10) = 144$, so $(x - 8)(x + 18) = 0$ and $x = 8$.

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Example 2.
In triangle ABC, the circle through A tangent to BC at B meets line AC again at P, and the circle through A tangent to BC at C meets line AB again at Q. Prove that the circumcentre of triangle APQ is equidistant from B and C.

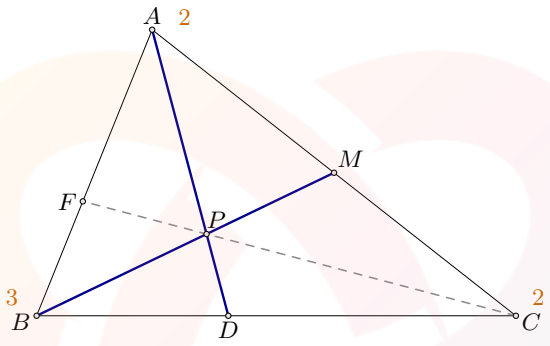
Solution.
You met this configuration in L4-GE1, where angles produced a fixed point. Angle chasing gives many equal angles but nothing about the circumcentre Z. Instead, if ρ is the radius of circle APQ, then $ZB^2 - \rho^2$ and $ZC^2 - \rho^2$ are the powers of B and C. So it suffices to show that B and C have equal power with respect to circle APQ.
B lies on line AQ, so its power is $\overline{BQ} \cdot \overline{BA}$. But Q and A also lie on the circle tangent to BC at C, and BC is a tangent from B to that circle, so $\overline{BQ} \cdot \overline{BA} = BC^2$. In the same way the power of C is $\overline{CP} \cdot \overline{CA} = CB^2$. The powers are equal, and it follows that $ZB = ZC$. Because the lengths are signed, no case analysis is needed: wherever P and Q fall on their lines, the products are the powers.



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Example 3.
In triangle ABC of area 21, $AB = 6$ and $AC = 9$. The bisector of angle A meets BC at D, the median from B meets AC at its midpoint M, and AD meets BM at P. Line CP meets AB at F. Find $AP : PD$, $BP : PM$, $AF : FB$ and the area of triangle APB.

Solution.
Coordinates would work but are slow, and BC is not given. The data are ratios on two sides, so we use mass points. By the angle bisector theorem $BD : DC = 6 : 9 = 2 : 3$, so we need $m_C : m_B = 2 : 3$; take $m_B = 3$, $m_C = 2$. As M is the midpoint of AC, $m_A = m_C = 2$.
Then $AP : PD = (m_B + m_C) : m_A = 5 : 2$, $BP : PM = (m_A + m_C) : m_B = 4 : 3$ and $AF : FB = m_B : m_A = 3 : 2$. For the area, $[APB] : [ABC] = m_C : (m_A + m_B + m_C) = 2 : 7$, so $[APB] = 6$. Check with Ceva: $\frac{2}{3} \cdot 1 \cdot \frac{3}{2} = 1$. A common error is to put $BD : DC = m_B : m_C$, which is the wrong way round.



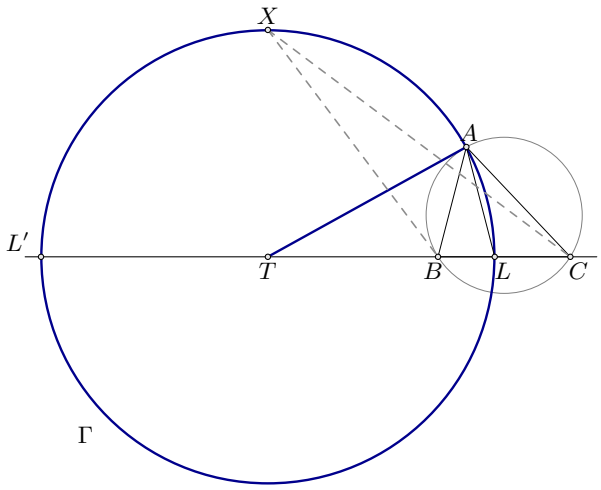
Example 4.
Triangle ABC has $AB = 6$, $AC = 8$, $BC = 7$. The tangent to its circumcircle at A meets line BC at T. (a) Find TA. (b) Prove that every point X on the circle Γ with centre T through A satisfies $XB : XC = 3 : 4$.

Solution.
(a) One could find angles with the cosine rule, but the tangent gives a similarity directly. Since $AB < AC$ we have $\angle C < \angle B$, and the tangent at A meets line BC beyond B. By the alternate segment theorem $\angle TAB = \angle ACB$, and triangles TAB, TCA share the angle at T, so they are similar and

$$\frac{TA}{TC} = \frac{TB}{TA} = \frac{AB}{CA} = \frac{3}{4}.$$

Thus $TB = \frac{9}{16}TC$ and $TC - TB = 7$. This gives $TC = 16$, $TB = 9$, and $TA = 12$; indeed $TA^2 = 144 = TB \cdot TC$.
(b) For X on Γ but not on line BC, $TX^2 = TA^2 = TB \cdot TC$, so by the converse of power of a point TX is tangent to circle XBC. Exactly as in (a), $\triangle TXB \sim \triangle TCX$, so $XB : XC = TX : TC = 12 : 16 = 3 : 4$. The two points of Γ on line BC are L with $BL = 3$, $LC = 4$, and L' with $L'B = 21$, $L'C = 28$, which also fit.
So Γ is the Apollonius circle of the ratio $3 : 4$. Note that L is the foot of the bisector from A, so $TA = TL$, a fact often used in AIME problems.

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Common traps.

- Forgetting signs: inside the circle, $PA \cdot PB = r^2 - OP^2$. Signed lengths handle every case at once.
- Putting mass points backwards: $BD : DC = m_C : m_B$.
- Applying Ceva to points on extensions, or Menelaus to points all on the sides, without care.
- Ratio lemma with the wrong angles: they are the angles at A, not at D.
- In AIME computations, keeping the wrong root of a quadratic or the wrong configuration (is T beyond B or beyond C?).

1.6 How to spot this in a contest

The usual signals are as follows.

1. Products of lengths along lines through one point: power of a point.
2. Two circles with a common chord or a common tangent: the radical axis. Three circles, or three lines that should be concurrent: the radical centre.
3. A tangent at a vertex meeting the opposite side: the similarity $TAB \sim TCA$.
4. Angles at a vertex together with a ratio on the opposite side: the ratio lemma or trig Ceva.
5. Cevians with given ratios, or areas cut by cevians: mass points.

In AIME problems, compute in exact fractions and check the answer against a rough scale drawing.

2 Practice problems

The problems are roughly in order of difficulty, from ⋆ to ⋆⋆⋆⋆. Hints are in Section 4 and full solutions in Section 5.

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Problem 1 ⋆ Circles ω_1 and ω_2 have radii 13 and 15, and their centres are 14 apart. A point P on the line through their two common points is at distance 20 from the line of centres. Find the length of a tangent from P to ω_1.

Problem 2 ⋆⋆ Triangle ABC has area 120. Point D lies on BC with $BD : DC = 1 : 2$, and E lies on CA with $CE : EA = 1 : 3$. The segments AD and BE meet at P. Find the area of quadrilateral $PDCE$.

Problem 3 ⋆⋆ In triangle ABC, $AB = 4$, $AC = 6$ and $\angle BAC = 120^\circ$. The point D on BC satisfies $\angle BAD = 90^\circ$. Find BD/DC.

Problem 4 ⋆⋆⋆ From a point P outside a circle ω with centre O, the tangents touch ω at A and B, and M is the midpoint of AB. A line through P, not through O, meets ω at C and D. Prove that O, M, C, D are concyclic.

Problem 5 ⋆⋆⋆ Triangle ABC has $AB = 13$, $BC = 14$, $CA = 15$. The circle with diameter BC meets AB again at F and AC again at E. Line EF meets line BC at T. Find TB.

Problem 6 ⋆⋆⋆⋆ Let ABC be an acute triangle with $AB \neq AC$, orthocentre H, and let M be the midpoint of BC. The circle through A tangent to BC at B and the circle through A tangent to BC at C meet again at K. Prove that K lies on AM and that $HK \perp AM$.

3 Contest ladder

Real past problems on the ideas of this unit, easiest first. Each source is a link to the official paper or the AoPS wiki; the note under it says where to start, not how to finish.

1. **AIME 1985 Problem 6** ⋆⋆
   Triangles with a common height have areas in the ratio of their bases.
2. **BMO1 2009/10 Problem 4** ⋆⋆
   Express everything through the two radii, and find the angle between DE and BC.
3. **AIME 1989 Problem 15** ⋆⋆⋆
   Mass points give every ratio on the cevians; then Stewart's theorem.
4. **BMO1 2007/08 Problem 5** ⋆⋆⋆
   Write each ratio in terms of the area weights of P, then use AM-GM.
5. **BMO1 2012/13 Problem 6** ⋆⋆⋆
   The tangencies give a pair of similar triangles sharing the vertex D.
6. **AIME II 2016 Problem 10** ⋆⋆⋆
   Ratio lemma (or similar triangles) twice, through the angles at C.
7. **USAJMO 2012 Problem 1** ⋆⋆⋆
   Read each angle condition as a tangency, then compare powers of points.
8. **BMO1 2025/26 Problem 4** ⋆⋆⋆
   Angles from the two tangencies give similar triangles with a common vertex at M.
9. **AIME I 2019 Problem 13** ⋆⋆⋆
   Find angle DFE first, then use the power of the point where CF meets line AB.

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10. **BMO1 2016/17 Problem 5** ⋆⋆⋆
Turn equal areas into equal lengths, then use the power of the point Q.

11. **AIME I 2016 Problem 15** ⋆⋆⋆⋆
Radical centre of the three circles, then similar triangles.

12. **AIME I 2019 Problem 15** ⋆⋆⋆⋆
Locate Q on line XY using the centre of the big circle, then power of P.

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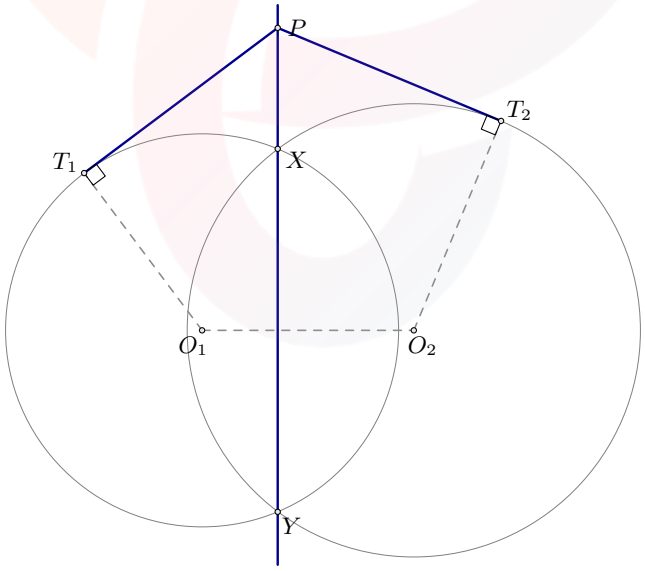
4 Hints

1. Locate the radical axis with the formula, then the power of P is $O_1P^2 - 13^2$.
2. Choose masses to match both ratios; the masses are proportional to the areas $[PBC]$, $[PCA]$, $[PAB]$.
3. You know both angles at A on either side of AD.
4. Express PA^2 in two ways: once with the circle, once with the right-angled triangle PAO.
5. Menelaus works. Alternatively, line EF is the radical axis of the circle on diameter BC and another circle through E and F that meets BC at known points.
6. For the first part, compare the powers of M. For the second, show K is on circle BHC, and find a diameter of that circle with one end on line AM.

5 Solutions

Problem 1

Circles ω_1 and ω_2 have radii 13 and 15, and their centres are 14 apart. A point P on the line through their two common points is at distance 20 from the line of centres. Find the length of a tangent from P to ω_1.



Put $O_1 = (0,0)$ and $O_2 = (14,0)$. The radical axis is the line $x = \dfrac{14^2 + 13^2 - 15^2}{2 \cdot 14} = 5$, so $P = (5,20)$ (or its mirror image, which gives the same answer). The square of the tangent length is the power $O_1P^2 - 13^2 = 25 + 400 - 169 = 256$, so the tangent has length 16.

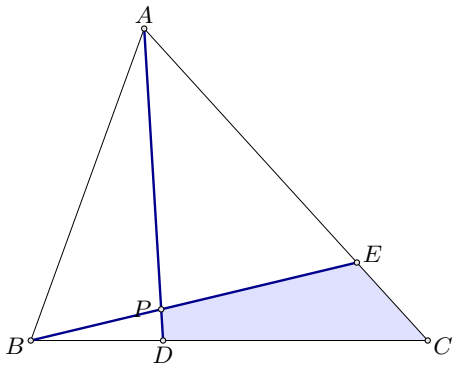
As a check, the power with respect to ω_2 is $9^2 + 20^2 - 15^2 = 256$ too, as it must be for a point on the radical axis.

Answer: 16

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Problem 2

Triangle ABC has area 120. Point D lies on BC with $BD : DC = 1 : 2$, and E lies on CA with $CE : EA = 1 : 3$. The segments AD and BE meet at P. Find the area of quadrilateral $PDCE$.



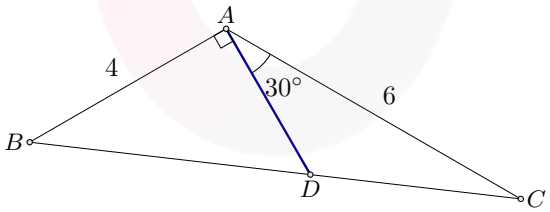
We need $m_C : m_B = BD : DC = 1 : 2$ and $m_A : m_C = CE : EA = 1 : 3$. Take $m_A = 1$, $m_C = 3$, $m_B = 6$, with total 10. Then P is the centre of mass and $[PBC] : [PCA] : [PAB] = 1 : 6 : 3$, so $[PBC] = 12$ and $[PCA] = 72$.

Triangles PDC and PBC share the height from P, so $[PDC] = \frac{DC}{BC}[PBC] = \frac{2}{3} \cdot 12 = 8$. Similarly $[PCE] = \frac{CE}{CA}[PCA] = \frac{1}{4} \cdot 72 = 18$. Hence $[PDCE] = 8 + 18 = 26$.

Answer: 26

Problem 3

In triangle ABC, $AB = 4$, $AC = 6$ and $\angle BAC = 120^\circ$. The point D on BC satisfies $\angle BAD = 90^\circ$. Find BD/DC.



Since $\angle BAD = 90^\circ$, $\angle DAC = 120^\circ - 90^\circ = 30^\circ$. By the ratio lemma,

$$\frac{BD}{DC} = \frac{AB \sin \angle BAD}{AC \sin \angle DAC} = \frac{4 \cdot 1}{6 \cdot \frac{1}{2}} = \frac{4}{3}.$$

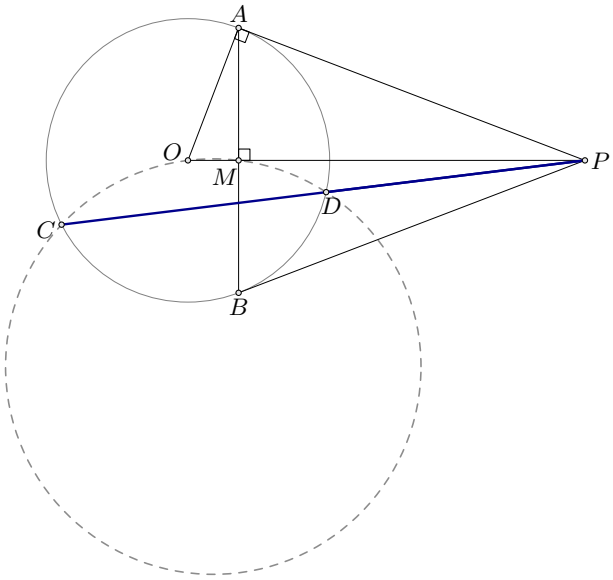
The length BC was not needed.

Answer: $\frac{4}{3}$

Problem 4

From a point P outside a circle ω with centre O, the tangents touch ω at A and B, and M is the midpoint of AB. A line through P, not through O, meets ω at C and D. Prove that O, M, C, D are concyclic.

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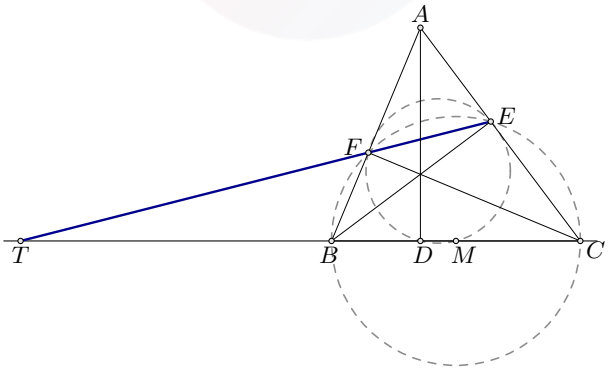


Since $PA = PB$ and $OA = OB$, both P and O lie on the perpendicular bisector of AB. So line PO is that bisector, it passes through the midpoint M, and $AM \perp PO$. The radius OA is perpendicular to the tangent PA, so triangle PAO has a right angle at A, and AM is its altitude. Triangles PMA and PAO are similar (they share the angle at P and each has a right angle), giving $PM \cdot PO = PA^2$.

By power of a point, $PA^2 = PC \cdot PD$. Therefore $PM \cdot PO = PC \cdot PD$. Here M and O lie on one ray from P, and C and D lie on another ray from P, since P is outside the circle, so the equality holds for signed lengths as well. The two lines are different because the line PCD does not pass through O. By the converse of power of a point, O, M, C, D are concyclic. ■

Problem 5

Triangle ABC has $AB = 13$, $BC = 14$, $CA = 15$. The circle with diameter BC meets AB again at F and AC again at E. Line EF meets line BC at T. Find TB.



The angles $\angle BFC$ and $\angle BEC$ are right, so F and E are the feet of the altitudes from C and B. Let D be the foot of the altitude from A and M the midpoint of BC; the familiar 13-14-15 split gives $BD = 5$, $DC = 9$, and $BM = 7$.

The nine-point circle passes through D, E, F, M, and the circle with diameter BC passes through B, C, E, F. Their radical axis is line EF, so T has equal powers: $\overline{TB} \cdot \overline{TC} = \overline{TD} \cdot \overline{TM}$. Since E and F

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lie on the sides, Menelaus says T is outside segment BC, and T is on the side of B (checked below). With $TB = x$,

$$x(x+14) = (x+5)(x+7),$$

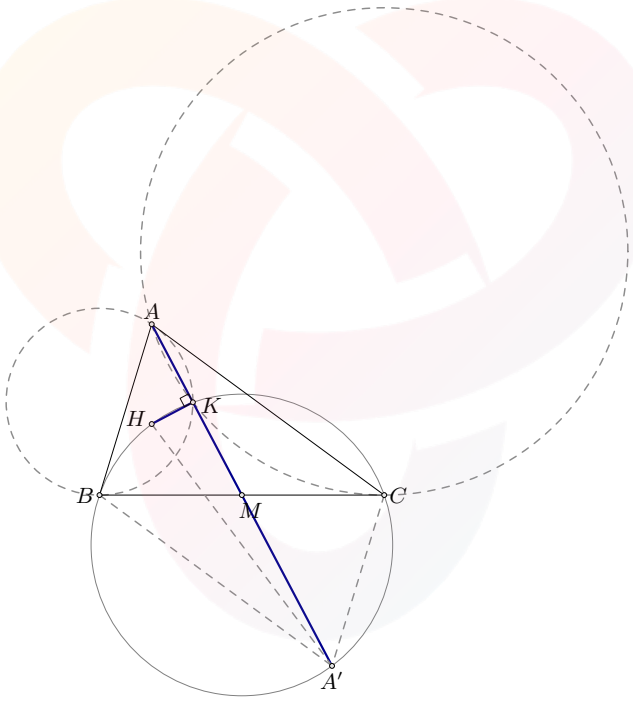
so $14x = 12x + 35$ and $x = \frac{35}{2}$.

Check by Menelaus: $BF = 14\cos B = \frac{70}{13}$, $AF = \frac{99}{13}$, $CE = 14\cos C = \frac{42}{5}$, $AE = \frac{33}{5}$, and $\frac{AF}{FB} \cdot \frac{BT}{TC} \cdot \frac{CE}{EA} = 1$ gives $\frac{BT}{TC} = \frac{70}{99} \cdot \frac{33}{42} = \frac{5}{9}$. So $TB < TC$, T is beyond B, and $TC - TB = 14$ gives $TB = \frac{35}{2}$ again.

Answer: $\frac{35}{2}$

Problem 6

Let ABC be an acute triangle with $AB \neq AC$, orthocentre H, and let M be the midpoint of BC. The circle through A tangent to BC at B and the circle through A tangent to BC at C meet again at K. Prove that K lies on AM and that $HK \perp AM$.



Call the circles Γ_B (tangent at B) and Γ_C (tangent at C).

K lies on AM. The power of M with respect to Γ_B is MB^2, since MB is a tangent, and with respect to Γ_C it is MC^2. As $MB = MC$, M is on the radical axis of Γ_B and Γ_C. So is A, a common point, and $A \neq M$, so the radical axis is line AM. The common point K also lies on it, so A, K, M are collinear. Moreover $\overline{MK} \cdot \overline{MA} = MB^2$. Since $\angle BAC$ is acute, A lies outside the circle with diameter BC, so $MA > MB$; hence $0 < MK < MA$ and K lies on segment MA, with $K \neq A$.

K lies on circle BHC. Use directed angles modulo $180°$. The tangent criterion in Γ_B gives $\measuredangle BKA = \measuredangle(BC, BA)$, and in Γ_C gives $\measuredangle CKA = \measuredangle(CB, CA)$. Hence

$$\measuredangle BKC = \measuredangle BKA - \measuredangle CKA = \measuredangle(BC, BA) + \measuredangle(CA, BC) = \measuredangle(CA, BA) = \measuredangle CAB.$$

Since $HB \perp AC$ and $HC \perp AB$, turning both lines through $90°$ gives $\measuredangle BHC = \measuredangle(AC, AB) = \measuredangle CAB$. So $\measuredangle BKC = \measuredangle BHC$ and B, K, H, C are concyclic.

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The right angle. Let A' be the reflection of A in M, so $ABA'C$ is a parallelogram. Then $BA' \parallel AC \perp BH$, so $\angle HBA' = 90°$, and likewise $\angle HCA' = 90°$. Hence A' lies on circle BHC and HA' is a diameter. Now $K \neq A'$: otherwise $MK = MA$ and $MK \cdot MA = MB^2$ would give $MA = MB$, so $\angle BAC = 90°$, which is false. Also $K \neq H$, since H on the median AM would make the altitude and the median from A coincide, forcing $AB = AC$. So K is a point of the circle with diameter HA' other than its ends, and $\angle HKA' = 90°$. As K and A' lie on line AM, $HK \perp AM$. ■